

Hide menu

D) Problem Overview

Video: 12 Problem Overview  
7 min

Graded Assignment: Assessment 13  
Submitted

Your grade: 95%

Your latest: 95% • Your highest: 95% • To pass you need at least 80% to receive your highest score.

## Assessment 13

Next item →

1. Which of the following best describes the primary objective of a Toffoli gate in quantum computing?

1 / 1 point

- ☒ It performs a controlled-controlled-NOT operation.
- ☐ It swaps the states of two qubits.
- ☐ It measures the state of a qubit to review what you've learned before starting the assignment? I'm here to help.
- ☐ It applies a phase shift to a qubit.

**Correct**

Correct! The Toffoli gate performs a controlled-controlled-NOT operation, which is a fundamental building block in quantum computing.

**Due**

Dec 25, 11:59 PM IST

**Attempts**

Unlimited

**Retry**

2. Which of the following statements is true about the Toffoli gate?

1 / 1 point

- ☒ It requires two control qubits and one target qubit.
- ☐ It is a single-qubit gate.
- ☒ It can be decomposed using Hadamard and CNOT gates.
- ☐ It can only be implemented on classical computers.

**Correct**

Correct! The Toffoli gate, also known as a controlled-controlled-NOT gate, requires two control qubits and one target qubit.

**Your grade**

95%

**View submission**

**See feedback**

3. Which circuit element is used to create entanglement between qubits?

1 / 1 point

- ☐ Hadamard Gate
- ☒ CNOT Gate
- ☐ Pauli-Z Gate
- ☐ Phase Gate

**Correct**

Correct! CNOT Gate is used to create entanglement between qubits.

4. Which of the following is a significant aspect of the Deutsch algorithm?

1 / 1 point

- ☐ It allows for faster-than-light communication.
- ☐ It requires no error correction mechanisms.
- ☐ It solves complex classical problems using quantum entanglement.
- ☒ It demonstrates the quantum parallelism feature.

**Correct**

Correct! The Deutsch algorithm showcases quantum parallelism, which is a fundamental concept in quantum computing.

5. Which of the following statements are true about balanced functions in quantum computing?

0.75 / 1 point

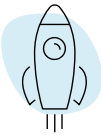
- ☐ They produce different outputs for different inputs.
- ☐ They can be determined with a single evaluation in classical computing.
- ☐ They are not relevant in the context of the Deutsch algorithm.
- ☒ They are the focus of the Deutsch-Jozsa algorithm.

**Correct**

Correct! The Deutsch-Jozsa algorithm is designed to detect balanced functions efficiently.

You didn't select all the correct answers

Close ×



You're ahead of the game!

Continue this momentum and you'll finish  
14 days earlier than expected.